

Coffee Shop Sales Dashboard - Detailed Case Study

Project Overview

This project demonstrates an end-to-end Business Intelligence solution built in Power BI using a realistic coffee shop sales dataset containing 5,000 sales transactions. The goal was to transform raw transactional data into an interactive dashboard that helps management monitor sales performance, profitability, customer behavior, and store performance.

Business Problem

Management requested a dashboard that identifies revenue drivers, underperforming areas, store performance, monthly trends, weekday patterns, and profitability to support strategic decision-making.

Dataset

The solution uses a star schema consisting of one FactSales table (5,000 rows) and dimension tables: DimProduct, DimStore, DimCustomer, and DimPromo. The fact table includes transaction date, time, quantity, revenue, profit, discounts, payment method, weather, sales channel, ratings, preparation time, and waiting time.

Data Preparation

Power Query was used for data cleaning, validation, datatype conversion, and loading. A Calendar table was created using DAX to enable time intelligence calculations.

Data Model

A star schema was implemented with FactSales connected to Product, Store, Customer, Promotion, and Calendar dimensions using one-to-many relationships.

DAX Development

Measures were organized into folders including Basic Measures, Previous Month (PM), Month-over-Month (MoM), Calculated Fields, and Insight measures. KPIs include Total Revenue, Total Profit, Total Transactions, Average Order Value, Profit Margin %, Revenue MoM %, Profit MoM %, Transaction MoM %, and dynamic insight messages.

Dashboard Features

Interactive slicers for Segment, Year, and Region allow users to filter results. Visuals include monthly revenue trend, store revenue comparison, KPI cards, product category revenue, store type comparison, and dynamic business insights.

Key Insights

Coffee products contribute the largest share of revenue. Urban stores generate significantly higher revenue than suburban stores. Revenue trends fluctuate throughout the year with noticeable monthly variations. Store-level analysis highlights top-performing and underperforming branches, enabling management to allocate resources effectively.

Business Value

The dashboard enables executives to monitor performance in real time, identify declining trends, optimize promotions, improve profitability, and make evidence-based business decisions.

Skills Demonstrated

Power BI, Power Query, Data Modeling, Star Schema, DAX, Time Intelligence, KPI Design, Interactive Dashboards, Business Storytelling, and Data Visualization.